1. Basics

① Def: the homotopy Lie algebra of R, $\pi^*(R)$, is:

$$\pi^*(R) := \pi^*(\mathrm{Quad}(A)).$$

where $A := Q[X]$, minimal model for R over Q, $\widehat{R} \cong Q/I$.

② Compute $\pi^n(R)$.

- By def., $\pi^n(R) := (\pi^*(R))^n$, degree n part.

  Then $\pi^n(R) = (kX_{n-1})^\vee$

  ($n-1$ since it's ΣkX.)

$\Rightarrow$ dual of a free v.s. gives ✡✡ $\dim_k(\pi^n(R)) = |X_{n-1}| = \varepsilon_n(R)$.

- Thus $\pi^n(R)$ is independent of min. model since $Q[X] \cong Q[Y]$ and $|X_{n-1}| = |Y_{n-1}|$.

  Moreover, $\pi^n(R)$ is independent of min. reg. repn., by Thm 2.2.12 in BEJ.

- Also, $\{x^\vee : x \in X_{n-1}\}$ forms a basis of $\pi_n(R)$.

e.g. $\{x_1, \cdots, x_e\}$ basis for $m/m^2 \Rightarrow \{x_1^\vee, \cdots, x_e^\vee\}$ basis for $\pi^1(R)$.

$\{f_1, \cdots, f_n\}$ basis for $I/mI \Rightarrow \{f_1^\vee, \cdots, f_n^\vee\}$ basis for $\pi^2(R)$.

③ Computing the brackets ✡✡

i) To compute brackets, we need to find their operations on basis: $[z^\vee, y^\vee]$ for $y, z \in X$.

To determine $[z^\vee, y^\vee] \in (\Sigma kX)^\vee$, we need their value on $x \in X$, basis for kX:

$[z^v, y^v](x)$.

We order $x, y, z \in X$ by a total order: $x < y$ if $|x| < |y|$.

Denote $|y^v| = |y| + 1$. Note that the actual degree of y^v is $|y|$, and $|y^v|$ is the shifted one.

ii) (See top of 65 for def. of pairing).

- To compute the pairing we need $\partial_q(x)$.

$$\partial_q(x) = \sum_{\substack{u \le v \in X \\ |u|+|v|+1 = |x|}} C_{x,u,v}\, uv, \quad C_{x,u,v} \in k.$$

- By def.,

$$[z^v, y^v](x) = \langle z^v y^v - (-1)^{|y^v||z^v|} y^v z^v, \sum_{u \le v} C_{x,u,v}\, uv \rangle$$

$$= \sum_{u \le v} (-1)^{|z|} C_{x,u,v}\, z^v(u)\, y^v(v)$$

$$- \sum_{u \le v} (-1)^{|y|+|y^v||z^v|} C_{x,u,v}\, y^v(u)\, z^v(v)$$

Observe that if $y < z$, then by $u \le v$ the only terms that survive is:

$$[z^v, y^v](x) = -(-1)^{|y|+|y^v||z^v|} C_{x,y,z}$$

Thus: $[y^v, z^v](x) = (-1)^{|y|} C_{x,y,z}$.

- Similarly, for $|y^v|$ odd,

$$(y^v)^{[2]}(x) = (-1)^{|y|} C_{x,y,y} = C_{x,y,y}.$$

☆☆ Note: for char $\neq 2$, $[y^v, y^v] = 2(y^v)^{[2]}$, and othw. $[y^v, y^v] = 0$.

- Result:

$[y^\vee, z^\vee] = (-1)^{|y|} \sum_{x \in X} c_{x,y,z} x^\vee$, $(y^\vee)^{[2]} = \sum_{x \in X} c_{x,y,y} x^\vee$ for $|y^\vee|$ odd ($|y|$ even).

$(yz)^\vee(u \cdot v) = y^\vee(u) z^\vee(v)$

Rmk: Note that $[y^\vee, z^\vee] = (\partial_q^\vee)(yz)^\vee$

$\partial_q^\vee$: dual of ∂_q : $\partial_q^\vee : (kx^2)^\vee \to (kx)^\vee$

Thus $[-,-]$ is det. by $\partial_q^\vee$.

iii) Computing $[\pi^1(R), \pi^1(R)]$.

- Let $m = (x_1, \cdots, x_e)$, $I = (f_1, \cdots, f_n)$.

  Thus $X_0 = \{x_1, \cdots, x_e\}$, $X_1 = \{y_1, \cdots, y_n\}$.

Then $\partial(y_i) = f_i \in m^2$ since Q/I min. reg. repn.

$\Rightarrow \partial(y_i) = f_i = \sum_{i \le j} c_{i,i,j} x_i x_j + w$, $w \in m^3$.

(m gen. by x_i, so m^2 gen. by $x_i x_j$.)

- Then $\partial_q(y_i) = \sum_{i \le j} c_{i,i,j} x_i x_j$

$\Rightarrow$ for $j < i$, $[x_i^\vee, x_j^\vee] = \sum_{l=1}^{n} c_{i,i,j} f_l^\vee$, $(x_i^\vee)^{[2]} = \sum_{l=1}^{n} c_{i,i,i} f_l^\vee$

($x_i^\vee$: degree 2, acts on X_1.)

Rmk: Here we identify generators of $\pi^2(R)$ with $f^\vee$, $f \in I$.

2. Examples.

① $Q = k[[x]]$, $R = Q/(x^2)$.

Then $Kos(x^2)$ is a minimal model for R:

$$0 \longrightarrow Q \xrightarrow{x^2} Q \longrightarrow 0.$$

- Thus $\pi^*(R) = (\bar{z} k x_0)^\vee \oplus (\bar{z} k x_1)^\vee = k x^\vee \oplus k y^\vee$, $\partial(y) = x^2$
  (gen. of m; "gen. of I").

- Thus the only nontrivial brackets are $[\pi_1(R), \pi_1(R)]$:
  Since $\partial(y) = f = 1 \cdot x^2$, the coefficient is 1.

$\Rightarrow (x^\vee)^{[2]} = 1 \cdot y^\vee = y^\vee$, and $[x^\vee, x^\vee] = 2y^\vee$ if $\mathrm{char}\, k \neq 2$.

② $Q = k[[x,y]]$, $R = Q/(f)$, $f = xy$.

Then $\pi^*(R) = (k x^\vee \oplus k y^\vee) \oplus k f^\vee$.

& $f = 1 \cdot xy \Rightarrow c_{1,x,y} = 1$

$$\Rightarrow \begin{cases} [x^\vee, y^\vee] = f^\vee \\ [y^\vee, x^\vee] = -(-1)^{1\cdot 1}[x^\vee, y^\vee] = f^\vee. \end{cases}$$

③ $Q = k[[x]]$, $R = Q/(f)$, $f = x^3$.

Then $\pi^*(R) = k x^\vee \oplus k f^\vee$, and $f = 0 \cdot x^2 + x^3 \Rightarrow$ trivial bracket.

④ More examples in 2.5.11 & 2.5.12 in BEJ.

3. C.I. & $\pi^*(R)$.

Thm: (R, m, k) local.

① R is regular $\Leftrightarrow \pi^*(R) = \pi^1(R)$

② R C.I. $\Leftrightarrow \pi^*(R) = \pi^1(R) \oplus \pi^2(R)$.

③ R not C.I., then $\pi^*(R)$ inf.-dim., and $\pi^n(R) \neq 0 \ \forall n \geq 1$.

pf: Note that $\dim_k(\pi^n(R)) = \varepsilon_n(R)$.

③ by Halperin's Rigidity.